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Crack suppression of glass welding by ultrafast laser without optical contact based on light modulation

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Outline

[Highlights](#)[Abstract](#)[Keywords](#)[1. Introduction](#)[2. Experiment](#)[3. Results](#)[4. Discussion](#)

Highlights

- The optical contact condition of glass welding was overcome by using ultrafast laser with ultra-high repetition rate.
- A method of glass welding with shaping beam was proposed. By shaping a circular Gaussian beam into an ellipse, the stress distribution can be changed to avoid the generation and propagation of microcracks.
- Experiments confirmed the successful crack-free welding of borosilicate glasses with a large gap.

Abstract

Glass is an excellent packaging material for microelectronic devices, and laser glass welding is a reliable method of achieving glass packaging. However, because glass is transparent to continuous or long pulse width laser, laser energy must be deposited directly through the nonlinear effect of ultrafast laser and glass. At the moment, ultrafast laser glass welding requires harsh optical contact conditions. Furthermore, because of the ultrafast laser's ultra-high peak power, the temperature within the laser action zone will rise and condense too quickly, causing the formation of thermal stress and microcracks, affecting the strength and stability of the weld and impeding the engineering application of this technology. In this paper, an ultrafast laser with an ultra-high repetition rate was used to increase nonlinear absorptivity and heat accumulation due to its short pulse interval, thereby increasing the size of the heat affected zone and liquid pool and enabling glass welding without optical contact. In addition, a slit was placed near the front of the objective lens to shape the incident circular Gaussian light into an elliptical Gaussian distribution, which could effectively inhibit the induction and generation of microcracks, ensure the forming quality of weld seams, and play an important role in promoting the engineering application of this technology.

Keywords

5. Conclusions

CRedit authorship contribution statement

Declaration of Competing Interest

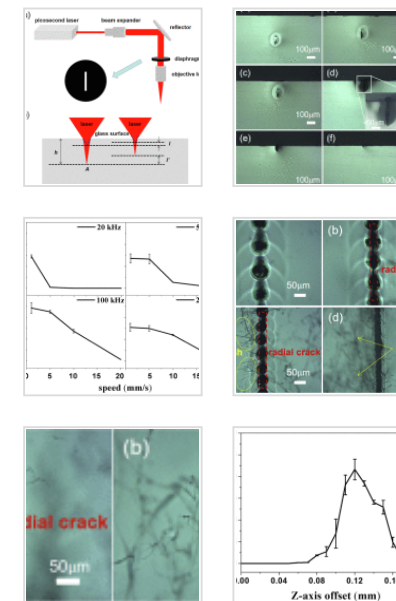
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Figures (9)



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Glass welding; Ultrafast laser; Crack suppression; Non-optical contact

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1. Introduction

Glass is a “neutral” material that is biocompatible and does not cause immune rejection. It also has a high hardness, strength, and chemical stability, as well as good light transmittance and a low thermal expansion coefficient [1], making it an excellent packaging material for solar cells, embedded microelectronics, organic light emitting diodes, micro electromechanical devices, micro sensors, and other electronic devices [2], [3]. The glass packaging must achieve a secure connection between the glasses. However, there are a number of issues and restrictions with the current approaches for connecting transparent brittle materials like glass in engineering since they are mostly based on silicon chip-related technologies [4], [5], [6]. Anodic bonding, for example, cannot be used for temperature or voltage sensitive electronic products because it requires high temperatures and pressure. Furthermore, if the connected materials do not have nearly the same thermal expansion coefficient, the equipment will not connect and may even break due to significant internal stress. Similarly, fusion welding, which is not a suitable procedure, calls for high temperatures and materials with similar thermal expansion coefficients. Another two examples are solid phase connections and adhesive bonding. The first necessitates a high surface quality and a lengthy welding time, whereas the second is hampered by thermal degradation and thermal expansion stress accumulation, neither of which is suitable for glass connection.

Due to the ultrafast laser's extraordinarily high peak power, melting and welding can be accomplished by directly depositing energy at the focus point via nonlinear effect [7]. It is an excellent solution for the glass packaging, which has recently been vigorously developed. Ultrafast laser has many advantages in glass welding. For starters, because the

action point is only near the focus, it has selective welding characteristics [8], [9]. Second, because of the multi-photon threshold effect and the Gaussian characteristics of the laser beam, the processed structure size can exceed the diffraction limit, allowing for high-precision micro welding. Furthermore, the interaction time between the ultrafast laser and the material is very short, which effectively avoids the thermal diffusion effect and does not result in high thermal stress due to thermal expansion coefficient differences [10], [11]. Finally, the weld is formed by melting its own material and has a high level of joint strength. However, it is precisely because of the ultrafast laser's extremely high peak power, the glass material will absorb a large amount of laser energy to reach a very high temperature in a very short time, forming a large temperature gradient, and then rapidly cool down after the laser pulse ends. Although this helps to avoid thermal diffusion effects, the temperature of the peripheral materials is always lower, limiting the expansion and condensation of the molten materials in the laser action zone, resulting in stress accumulation. Once the internal tensile stress formed exceeds the material's tensile threshold, microcracks form [12]. Currently, there is little research on the formation and suppression of microcracks. Isamu Miyamoto et al. [12] concluded that high repetition rate helps to inhibit the generation of microcracks when the laser acts on the interior of the glass material based on the heat conduction model and experimental verification. When the liquid pool area is not completely embedded in the material, the material's pressure restriction is weakened, and cracks still occur. It is also a significant reason for the optical contact required for glass welding, which can limit thermal stress release. Furthermore, through time modulation of pulse energy, Akinao Nakamura et al. [13] effectively avoided stress concentration and made it uniformly distributed, increasing the laser energy threshold for generating microcracks and helping to improve weld fracture strength.

The crack suppression methods described above are based on optical contact conditions that are extremely stringent, requiring the gap between glasses to be $1/4$ wavelength [7], or even 100nm [14]. Although this is helpful in limiting thermal stress release, it is insufficient for engineering applications. It is difficult to ensure the establishment of this

condition, especially for large area welding. As a result, in recent years, researchers have begun to investigate glass welding without optical contact and have proposed some efficient methods [14], [15], [16], [17]. However, how to suppress the generation of microcracks in such welding situations without good material or gap restriction but open interfaces has not yet been reported, which is critical for the engineering application of ultrafast laser glass welding.

To address the aforementioned issues, we proposed an ultrafast laser glass welding method with ultra-high repetition rate based on slit shaping in this paper. On the one hand, by increasing nonlinear absorptivity and heat accumulation, we could achieve a larger laser action zone and obtain more molten materials, thereby overcoming the optical contact conditions. Simultaneously, by reshaping the circular Gaussian beam into an ellipse, the distribution of the laser induced stress field was reshaped to avoid the generation and expansion of microcracks, greatly reducing the likelihood of weld failure. It was beneficial to promote the engineering application of ultrafast laser glass welding.

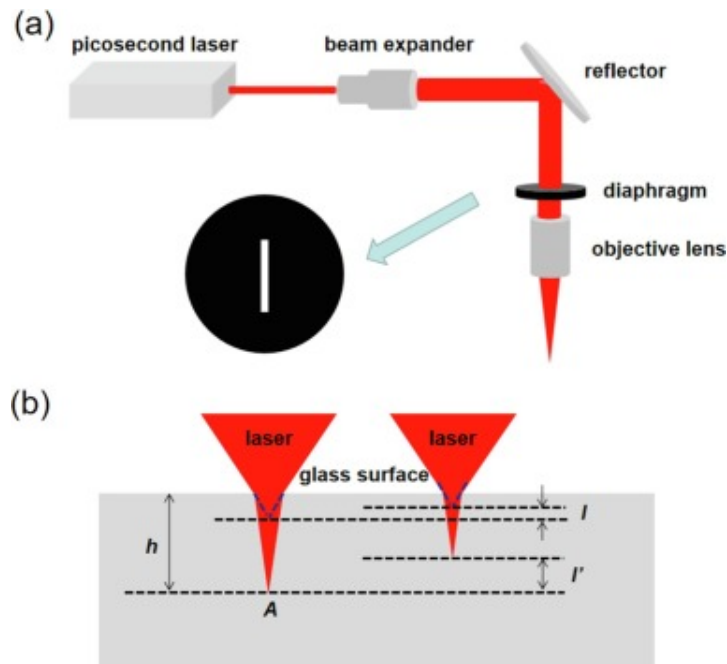
2. Experiment

2.1. Experimental equipment and principle

The processing light source in this experiment was a Nd:YVO₄ picosecond laser from Edgewave, which emitted 10ps pulses of 1064nm light with an exit spot size of about 2mm. The beam was then focused using a 3×beam expansion and a 20×objective lens (Mitutoyo NIR PLAN APO, NA0.42) to produce a focal spot of 5μm diameter. In order to change the energy distribution of the light field and improve the processing quality, a 1mm×10mm slit was set in front of the objective lens. The focal depth of the shaped beam was similar to that of the unshaped beam, which was about ±5μm. The major axis of the diffraction elliptical spot just passing through the slit was consistent with the major axis direction of the slit. After passing through the objective lens, the beam was still tightly focused in the major axis direction, while expanded R_x/R_y times in the minor axis[18]. Here R_x was the radius along the major axis of the elliptical beam and R_y was

that along the minor axis. As a result, the major axis of the elliptical focus spot was now 90 degrees off from its original orientation and perpendicular to the major axis of the slit.

Fig. 1(a) depicts the system's schematic diagram.



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Fig. 1. Diagram of the experimental platform and method. (a) experimental setup, (b) focus moving distance and position.

To weld borosilicate glasses (20mm×20mm×2mm), the laser fundamental frequency mode (50MHz) was used. Its extremely short pulse time interval could effectively increase heat accumulation and nonlinear absorption of the material to laser energy [16], [19], [20], [21], allowing the laser heat affected zone to expand. Additionally, the internal gate of the laser was a crucial parameter, which represented the gate signal width selected from the fundamental frequency, and the time interval between each gate signal was fixed to 10μs. In other words, the internal gate directly controlled the pulse

envelope's pulse count, which had an impact on heat accumulation. Simultaneously, the slit was used to shape the incident beam and adjust the energy distribution of the light field, thereby regulating the stress field and preventing the formation of microcracks.

2.2. Experimental process

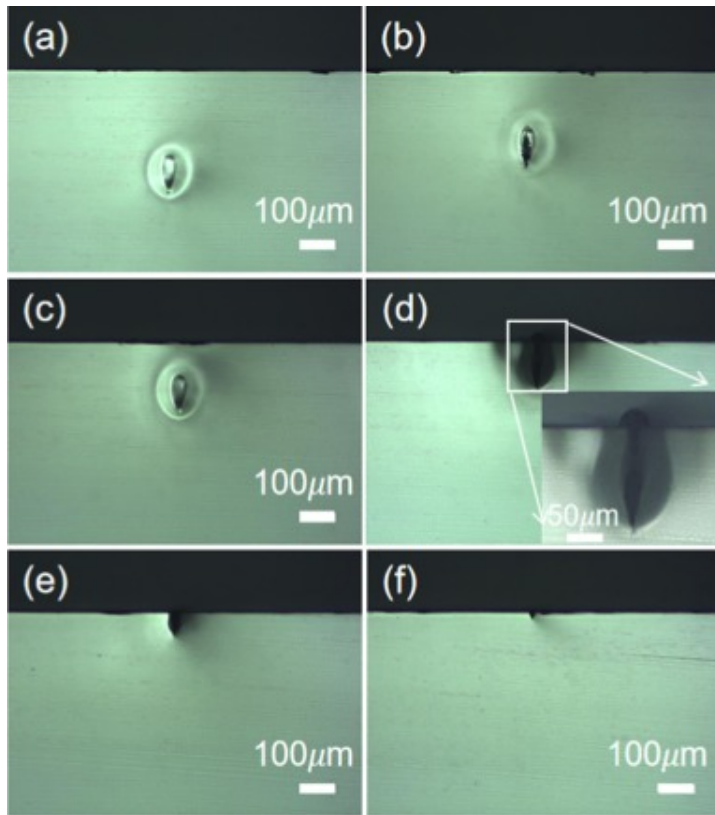
Initially, using the glass surface as the Z-axis zero point, focused a 2.3W laser with a internal gate of 50kHz on the point A in a single piece of glass, where the distance h from the zero point was 0.4mm, and driven the laser at a speed of 5mm/s for scribing processing. After that, gradually adjusted the Z-axis by 50 μ m to bring the focus closer to the glass surface, and kept scribing with the same parameters at various focus points. It is important to keep in mind that the moving distance l of the Z-axis is less than the actual moving distance l' of the focus in the glass due to the influence of refractive index, as shown in Fig. 1(b).

A microscope was used to examine the heat-affected zone and the bulge formed in the cross section after grinding. The focus position of bulge appeared was then locked, and scribing was resumed with a different internal gate and processing speed. By measuring the bulge height with a confocal microscope, the influence of laser parameters on the bulge height was discussed, and the optimization result of obtaining the maximum bulge height was obtained. In order to investigate the inhibition of the shaping beam on crack formation, the incident laser light field was then modulated by positioning a 1mm $\times$ 10mm slit in front of the objective lens [18], as shown in Fig. 1(a), and focusing the focus on various depths in the glass to observe crack formation on the glass surface with the optimized parameters. During the process, the laser output power must be increased in order to maintain the 2.3W of power behind the objective lens because the slit blocks 85% of the light. Finally, the shaping laser was used to weld naturally stacked glasses, and the shear strength of weld was tested by the universal material testing machine. In this process, in order to improve the accuracy of the test as much as possible, the sample was welded with multiple seams, and the average value of the test group of samples was taken.

3. Results

3.1. Parameter optimization without slit aperture

To move the focus towards the glass surface, adjust the Z-axis in 50 μm steps, and Fig. 2 shows the results of the heat affected zone formed by laser scribing inside the glass at 2.3W laser power, 50kHz internal gate, and 5mm/s scanning speed. When the focus is 0.4mm below the glass surface, as shown in Fig. 2(a), it can be seen that the length of the heat affected zone can reach about 200 μm under the condition of 50MHz, which is significantly better than the results obtained from previous studies using a high repetition rate below 10MHz [22], and can further increase the generated melt. The laser action zone similarly goes toward the glass surface without altering in size as the focus moves to the surface (Fig. 2(b) and (c) illustrate the cases where the Z-axis moves up by 50 μm and 100 μm , respectively). When the Z-axis moves up 150 μm (as shown in Fig. 2(d)), an obvious bulge appeared with a height of about 20 μm , indicating that the generated melt is sufficient to fill a large glass gap and overcome the optical contact condition. As the focus was moved upward, the bulge disappeared and the heat-affected zone was no longer visible in the cross section, only the groove created by the etching effect (as shown in Fig. 2(e)). The width of the laser etching groove reaches the minimum value when the Z-axis moves up 250 μm (as shown in Fig. 2(f)), indicating that the focus is close to the glass surface. The actual distance the focal point moves is greater than the distance the Z-axis moves due to the refractive index, as mentioned in the previous section, which is why the focal point reaches the glass surface when the Z-axis moves only 250 μm upward.



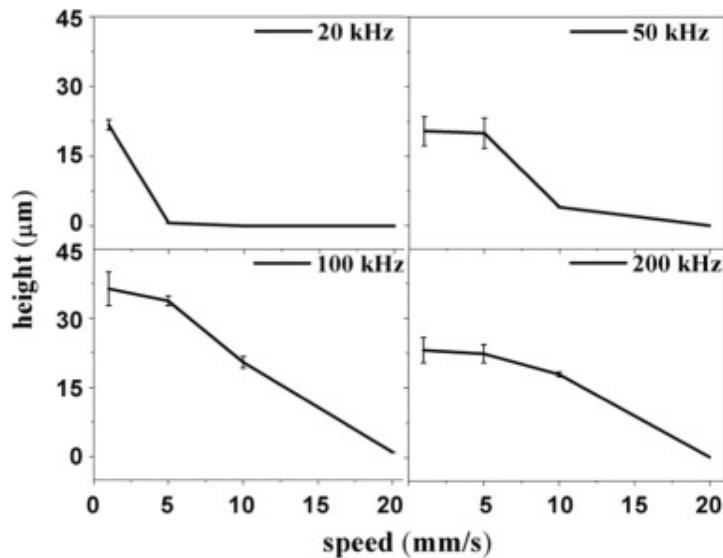
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Fig. 2. The state of the laser action zone where the focus moves from the inside of the glass to the surface at 2.3W laser power, 50kHz internal gate, and 5mm/s scanning speed. (a) The focus is 0.4mm below the glass surface, (b)-(f) adjust the Z-axis in 50μm steps to move the focus towards the surface in turn.

Keeping the focus position of the bulge appeared (the Z-axis moves up 150μm, Fig. 2(d)) unchanged, the results of adjusting the laser scanning speed and internal gate to investigate the influence of laser parameters on the molten bulge height on the glass surface are shown in Fig. 3. First of all, the bulge height decreases with the increase of scanning speed, as expected, because the superposition and accumulation of pulse energy

decreases. Secondly, as the internal gate frequency increases, the bulge height increases and then decreases, reaching a maximum at 100kHz. When the processing speed is 1 mm/s, the maximum bulge height is around 36 μ m, and the processing parameters are optimal at this time.



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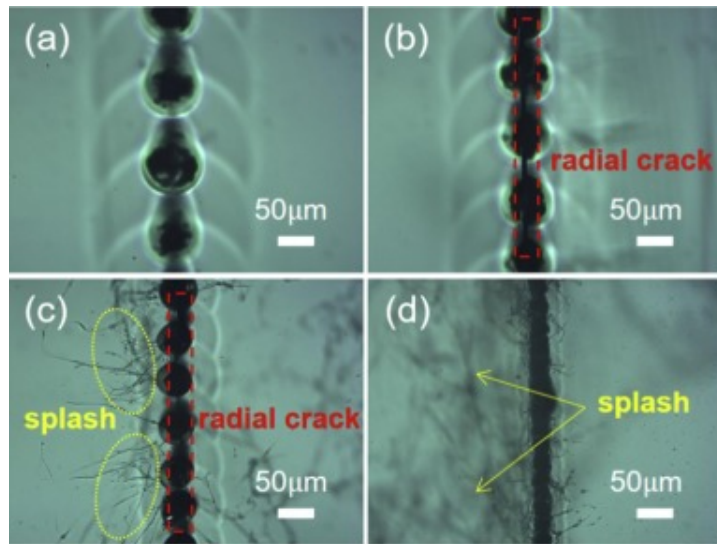
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Fig. 3. Variation of molten bulge height with laser scanning speed and internal gate.

3.2. Crack suppression based on a slit

A large bulge height was obtained through parameter optimization, which was greater than the glass gap of natural stacking [17], indicating that this welding method can achieve glass welding with a large gap. However, without the interface limitation, thermal stress will be released in the welding gap, resulting in the formation of microcracks. Fig. 4 depicts the observation of a glass surface using optimized parameters to focus on different depths in the glass. For recording purposes, the focus position is still represented by the upward distance of the Z-axis from -0.4mm. Wherein, Fig. 4 (a) depicts the situation when the Z-axis moved up 70 μ m and the liquid pool just rose to the glass

surface, forming a bulge. Prior to that, although the glass surface will have bulges as well, it is the material bulge in the self caused by the expansion of the laser action zone that will be visible. As the focus moves up, the liquid pool area more thoroughly penetrated the glass' surface, releasing heat pressure that led to the formation of microcracks and radial fissures that extended in the direction of the laser's processing, as depicted in Fig. 4 (b). Since then, the liquid pool area had been constantly moved out of the glass, and more molten materials protruded from the glass surface. At the same time, some molten materials were splashed due to thermal pressure, as shown in Fig. 4 (c). When the Z-axis moved up 210 μm , as shown in Fig. 4 (d), the liquid pool area had essentially moved out of the material, and the molten matter had been ejected without the limitation of the interface, so no effective bulge could form. Furthermore, because of the complete release of stress, microcracks no longer occurred, and the processing effect shifted from melting to etching. In addition, as shown in Fig. 4, the period of the surface bulge gradually decreases as the focus moves up. This is due to the fact that once the molten pool area breaks through the glass surface, the plasma rapidly dissipates in the free space, as does the shielding effect on subsequent lasers, and the laser returns to the original focus to begin the next cycle. This period is obviously related to the time when the laser begins to break through the surface from the focus inside the glass. As a result, as the focus moves up, the period for the laser to form a bulge on the glass surface gets shorter and shorter until they are connected.



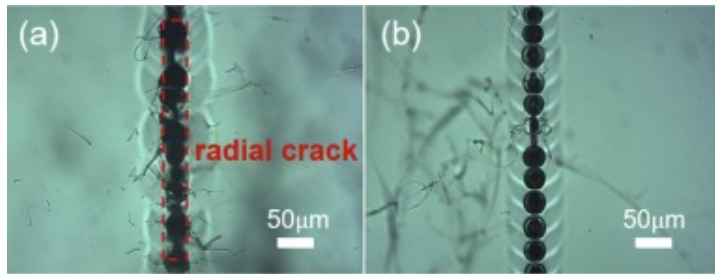
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Fig. 4. Results of laser action zone on glass surface with laser power of 2.3W, internal gate of 100kHz, scanning speed of 1 mm/s. (a) The Z-axis moves up 70μm from -0.4mm, (b) The Z-axis moves up 90μm from -0.4mm, (c) The Z-axis moves up 150μm from -0.4mm, (d) The Z-axis moves up 210μm from -0.4mm.

In order to suppress the generation of this radial crack, a 1 mm×10mm slit aperture was used to shape the incident laser and transform the circular Gaussian spot into an elliptical Gaussian spot distribution, and Fig. 5 shows the results when the Z-axis is moved 150μm up from -0.4mm. Wherein, Fig. 5 (a) shows that the major axis direction of the slit is perpendicular to the path, and Fig. 5 (b) shows that the major axis direction of the slit is parallel to the path. It is clear that when the major axis of the slit is perpendicular to the path, there are still obvious radial cracks, whereas when the major axis is parallel to the path, the cracks are effectively suppressed. Although there are also splashes around the processing area, the bulges in the liquid pool area under laser action appear to have distinct boundaries with each other. This is due to the suppression of microcracks and the

lack of a scattering effect on the light observed by the microscope, resulting in such a difference in results.

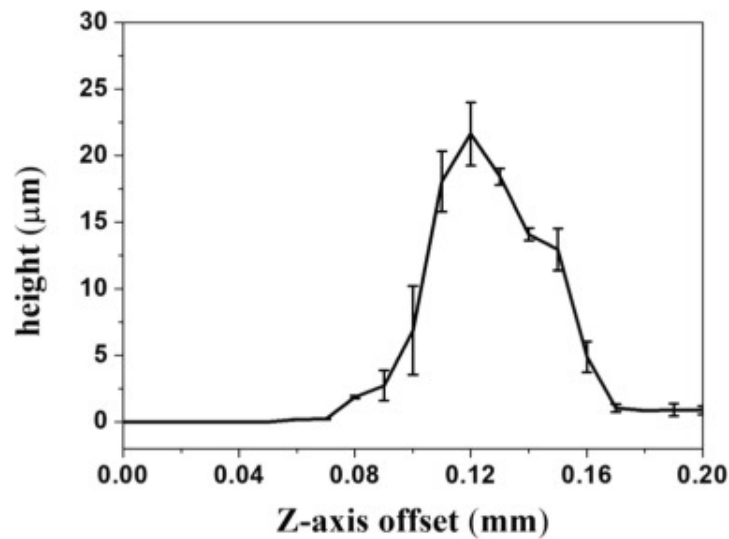


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Fig. 5. Effect of slit aperture on crack formation when the Z-axis is moved 150µm up from -0.4mm. (a) the major axis direction of the slit is perpendicular to the path, (b) the major axis direction of the slit is parallel to the path.

It should be noted that after adding a slit with the major axis direction parallel to the path, not only does the focus position shown in Fig. 5 not generate microcracks, but the experiments also show that other focus positions achieve microcrack suppression. The surface bulge height was measured at various focus offset positions, and the results are shown in Fig. 6, indicating that the maximum bulge height is about 21 µm. Although lower than in the case without the slit, it is adequate for the glass welding gap.



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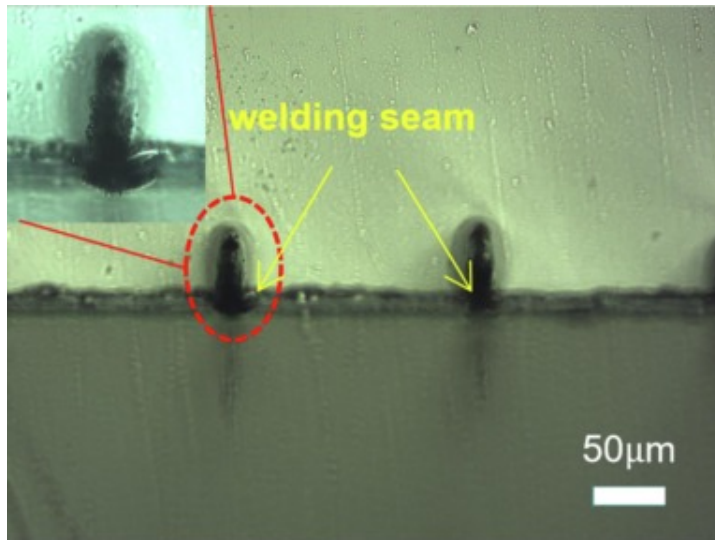
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Fig. 6. The relationship between the offset of the Z-axis from -0.4mm and the height of the liquid pool bulge after adding a slit whose major axis direction is parallel to the path.

3.3. Glass welding based on slit shaping

Using the optimized laser parameters to weld two pieces of naturally stacked borosilicate glass (without any external force) in the form of line welding, according to the crack suppression conditions obtained in the previous experiment, the results showed that the welding under this condition could be realized successfully. The gap between glasses is approximately $9.82\mu\text{m}$ after measurement. And, as shown in Fig. 7, the laser action zone covers the upper and lower glass layers, and enough molten materials are filled in the glass gap to form a weld, demonstrating that this welding method can completely overcome the optical contact condition. Furthermore, no microcracks are found in the weld zone, confirming the effectiveness of the crack suppression method. Finally, the universal material testing machine was used to test the weld shear strength. All samples

were welded successfully, with an average shear strength of 39MPa, indicating the reliability of the welding strength.



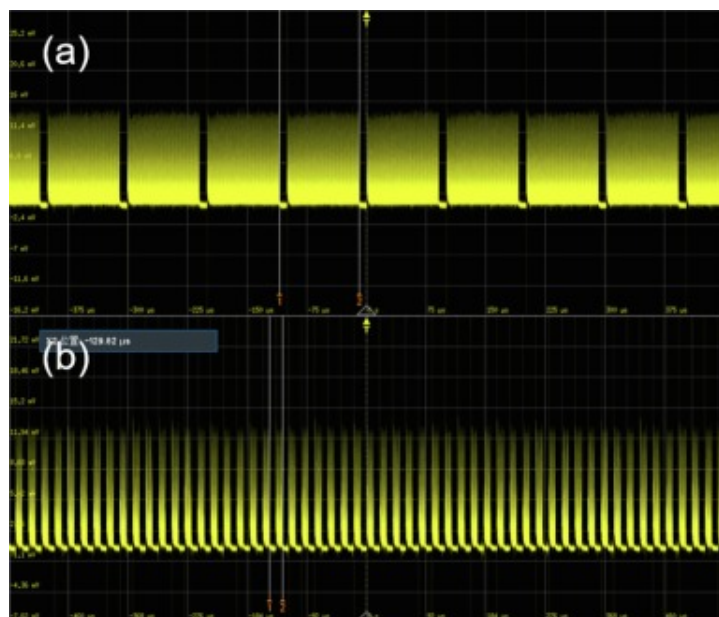
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Fig. 7. Welding results of shaping laser on naturally stacked borosilicate glasses.

4. Discussion

Because the fundamental mode of the Edgewave laser's repetition rate is fixed at 50MHz, the internal gate has a significant influence on modulated pulse energy and thermal accumulation as a frequency selective gate signal. The oscilloscope measurement results of the internal gate signal are shown in [Fig. 8](#), and it is clear that this parameter represents the gate signal width selected from the fundamental frequency. The time interval between each gate signal is fixed at 10μs, and the laser pulse output is equivalent to burst mode. Because of the extremely short pulse time interval (20ns) in burst mode, subsequent pulses can realize secondary excitation of defect state and local state electrons, resulting in higher nonlinear absorption [\[21\]](#).



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Fig. 8. Oscilloscope measurement results of internal gate signal. (a)10kHz, (b)50kHz.

The greater the internal gate value, the narrower the frequency selection gate signal width, the fewer pulses are accumulated, and the burst mode effect is weakened. Simultaneously, as the total number of pulses output per unit time decreases, the modulated pulse energy increases. Both of them play opposing roles in the formation of the laser-induced heat affected zone, resulting in a non-monotonic result, as illustrated in Fig. 3. When the internal gate is exactly at 100kHz, the laser modulated pulse energy and pulse cumulative superimposed heating effect reach their maximum value, resulting in the largest generated heat affected zone.

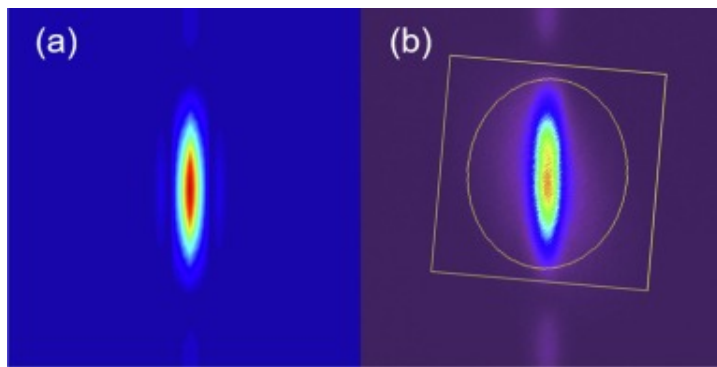
The interaction between the ultrafast laser and the glass material will deposit a large amount of energy in a short period of time, causing it to warm up. Following the pulse action, the massive temperature gradient will rapidly cool the action area. This process

will cause stress in the laser action zone, with thermal expansion represented by compressive stress and cooling contraction represented by tensile stress. When the action zone is located inside the material, the material limits its expansion process, resulting in a relatively weak contraction tensile stress during cooling. However, as the focus gradually moves to the sample surface until the liquid pool area breaks through the surface, the material's expansion limit on the action zone is weakened, and the melt can expand sufficiently at the free interface, inducing large shrinkage tensile stress during the cooling process. Because the tensile stress of brittle materials, such as glass, is far less than the compressive stress, microcracks are easily formed [23].

Because of the isotropic thermal effects in all plane directions, the naturally induced stress distribution is also isotropic for general circular Gaussian facula. If there is only one action point, microcracks will form when the tensile stress generated exceeds the material's tensile threshold. The crack generation direction is also random in all directions due to the isotropy of the stress distribution. Of course, there will be no microcracks if the tensile stress generated is less than the material's tensile threshold. If there are multiple laser action points close together, the stress in the direction of the path connected by the laser action points will form a resultant force. Even though the stress in a single action point is insufficient to generate microcracks, the resulting force may generate radial cracks, which is why there are radial cracks in Fig. 4 (b) (c) but no other microcracks around the action point.

Slit shaping is used in this paper to transform circular Gaussian facula into elliptical Gaussian facula. In Fig. 9, the simulated facula image in MATLAB (Fig. 9 (a)) and measured result (Fig. 9 (b)) are shown, with the major axis direction perpendicular to the slit's major axis direction. The distribution of the light field changes the distribution of the stress field, for an elliptical light spot, the stress distribution is concentrated at the tip on both sides of its major axis [24], [25], implying a condition for directional induction of microcracks. The majority of the stresses are still distributed on the path when the elliptical light spot's major axis is in that direction. It generates more resultant force in

comparison to the circular light spot, making it simpler to cause the development of radial cracks, as shown in Fig. 5 (a).



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Fig. 9. Result diagram of beam shaping. (a) Focused spot after slit shaping simulated by MATLAB, (b) measured image of slit shaping focus spot with major axis of $17\mu\text{m}$ and short axis of $4.7\mu\text{m}$.

When the major axis of the elliptical light spot is perpendicular to the path's direction, the laser's stress concentration area cannot form a resultant force, which greatly reduces the tensile stress upward of the path, improves the laser energy threshold that induces cracks in the material, aids in the inhibition of microcrack generation, and achieves better welding effects, as shown in Fig. 5 (b).

5. Conclusions

In this paper, a 50MHz ultra-high repetition rate ultra-fast laser combined with a slit aperture was used, which not only effectively increased the laser heat-affected zone and realized glass welding under natural stacking condition without external force constraints, but also shaped the incident Gaussian laser into an elliptical focus through the slit so that the stress was concentrated in the tip area. Then, by making the elliptical

light spot's major axis direction perpendicular to the path direction, the stress concentration area of each laser action zone was dispersed, and no resultant force could be formed, greatly reducing tensile stress and effectively inhibiting the generation of microcracks, and had important guiding significance for the engineering application of ultrafast laser glass welding.

CRediT authorship contribution statement

Hang Chen: Conceptualization, Methodology, Validation, Data curation, Writing – original draft. **Youmin Rong:** Software, Investigation, Writing – original draft. **Yu Huang:** Formal analysis, Visualization, Supervision, Funding acquisition. **Congyi Wu:** Resources, Writing – review & editing, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Data availability

Data will be made available on request.

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...Shen et al. [16] proposed a method that combined laser welding with ultrafast laser welding to achieve a connection between glass and metal with a gap in the range of hundreds of micrometers. Wu et al. [17] placed a slit near the front of the objective lens to shape the incident circular Gaussian light into an elliptical Gaussian distribution, which could effectively inhibit the induction and generation of microcracks. Zhang et al. [18] proposed that mJ-level pulsed femtosecond laser transmission welding can enhance joint strength, eliminate the need for special surface preparations, and potentially overcome the challenge of aluminum-fused silica welding with a large interface gap....

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...Based on this result, Takayuki Tamaki [5] introduced ultrafast laser into the welding of glasses for the first time, and observed the realization of direct welding under optical contact condition, pioneering the direct welding method for transparent brittle materials. Following that, researchers continued to conduct extensive researches on enhancing weld strength [6–14], micro mechanism processes [15–18], suppression of laser induced microcracks [19–21], and glass welding methods without optical contact [22–26]. However, due to the plasma shielding phenomenon and thermal diffusion process, the laser forms a double structure laser action zone with the shape of a water drop in the glass material [27]....

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